ARXA

Industrial Telemetry & Real-Options Operating Engine

Quantitative Staged Capital Allocation for Mission-Critical Production Assets

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Live Case Study | GitHub Repository
Standards: ISO 13374 | Columbia FE

Executive Summary & Strategic Context

Industrial process plants, energy utilities, and heavy manufacturing facilities face a systemic capital budgeting dilemma: predictive condition monitoring tools flood operators with statistical anomaly alerts, yet fail to evaluate whether executing a €50,000 turbine overhaul maximizes asset Net Present Value (NPV) under uncertainty. Calendar-based preventative maintenance discards remaining asset life prematurely, while running equipment to failure causes catastrophic downtime losses exceeding 10× to 30× the baseline repair cost.

ARXA was engineered at **Arxa Group** to unite physical IoT telemetry with quantitative financial engineering. ARXA enforces the operational axiom: *prediction has value only through a feasible action*. Combining sensor ingestion, statistical Remaining Useful Life (RUL) estimation, and a **Columbia-modeled Cox-Ross-Rubinstein (CRR) binomial real-options lattice**, ARXA evaluates the managerial option to *act* (intervene immediately) versus *defer* (preserve capital while gathering additional degradation telemetry).

System Architecture & Quantitative Engine

ARXA coordinates physical reliability engineering and financial optimization across five layers:

1. **High-Frequency Telemetry Ingestion:** Continuously cleans and extracts features from multi-channel vibration FFT spectral density, thermal decay gradients (℃), and dynamic rotational velocity (RPM) in compliance with ISO 13374. Guarantees zero lookahead bias via causal time-series windows.
2. **Statistical Degradation & RUL Modeling:** Fits continuous Weibull hazard functions and Arrhenius thermal degradation kinetics to compute residual equipment lifespan. Demonstrates verified stability: rolling Mean Absolute Error (MAE) of **0.039256** under nominal conditions and **0.039728** under noisy sensor dropouts.
3. **American-Style Binomial Real-Options Lattice:** Adapts financial engineering methodologies (Cox-Ross-Rubinstein discrete-time lattices) to value managerial flexibility. Computes optimal backward induction stopping rules:

$$V(S,t) = \max\left(S - K,\ e^{-r\Delta t}[pV_u + (1-p)V_d]\right)$$

   Where $S - K$ represents immediate irreversible maintenance CapEx and continuation value reflects deferral flexibility.
4. **Operational Feasibility & Resource Arbiter:** Evaluates physical crew shift availability, spare parts procurement lead times, and scheduled outage windows before triggering dispatch.
5. **Automated Dispatch & Telemetry Feedback:** Dispatches verified maintenance work orders to plant ERP/CMMS systems, continuously logging operator overrides and post-intervention telemetry.

5-Dimensional Multidimensional Operating Model

In accordance with Arxa Group Systems Architecture Standards, ARXA integrates five operational dimensions:

- **Technical Architecture:** Python time-series engine with rolling evaluation windows, OpenAPI 3.1 REST contracts, and isolated SQLite state ledgers.
- **Operating Workflow (RACI):** Plant Director (sets hurdle rates and downtime cost parameters) → Reliability Engineer (monitors telemetry calibration and RUL drift) → Maintenance Planner (certifies spare parts and crew shifts) → Control Room Operator (holds manual override authority).
- **Financial Model & Lifecycle Economics:** Evaluates 80 operational cycles under unmanaged vs. real-options regimes. Reduces total lifecycle costs from €540,000 (unmanaged baseline) to **€484,000** (real-options optimum), delivering **€56,000 in net cash savings** and completely eliminating catastrophic failures.
- **Risk & Governance Model:** Safety-critical interlocking logic: models pause automatically upon sensor dropouts, preventing uncalibrated automated decisions. Full compliance with ISO 17359 asset diagnostics.
- **Scaling & Integration:** Modular architecture engineered for edge micro-controller deployment (NVIDIA Jetson / Siemens IPC) and enterprise SCADA / SAP PM integration.

Empirical Sensitivity & Capacity Matrix

ARXA stress-tests 12 distinct operating regimes across varying intervention efficacy and maintenance crew capacity:

Efficacy	Capacity	Lifecycle Cost	Actions	Failures	Operational Regime Interpretation
0% (None)	0 actions	€540,000	0	6	Unmanaged baseline: 6 catastrophic failures.
0% (None)	Unlimited	€1,024,000	22	6	Worst-case: wasted €484k CapEx without preventing failure.
50% (Partial)	10 actions	€670,000	10	5	Partial efficacy reduces failures; cost bounded by crew capacity.
100% (Full)	10 actions	€580,000	10	4	Budget-constrained: reduces failures under strict CapEx limits.
100% (Full)	**Unlimited**	**€484,000**	**22**	**0**	**Full Real-Options Optimum: Zero failures; €56k net savings.**

Architecture Decision Records (ADRs) & API Contracts

All architectural trade-offs are formally governed by RFC-style Architecture Decision Records:

- **ADR-0001:** Strict Temporal Isolation & Forward-Rolling Evaluation Windows in Industrial IoT.
- **ADR-0002:** Cox-Ross-Rubinstein Binomial Lattices for Maintenance Real Options under Uncertainty.
- **ADR-0003:** Operational Capacity Constraints & 12-Regime Efficacy Sensitivity Enforcement.
- **OpenAPI 3.1 Contract:** Endpoints for ‘/v1/telemetry/stream‘, ‘/v1/rul/forecast‘, and ‘/v1/options/evaluate-deferral‘.

Methodological & Confidentiality Notice: This architectural whitepaper utilizes validated synthetic benchmark datasets and open institutional reliability frameworks (ISO 13374, ISO 17359). No confidential corporate banking or private client data from active or past institutional engagements (BNP Paribas CIB, Santander CIB) is contained herein. © 2026 Igor Grybek. All rights reserved.